



Institute of Technology of Cambodia

Department of Applied Mathematics and Statistics



Exploratory Data Analysis (I5-Project)

“Customer Segmentation Analysis”

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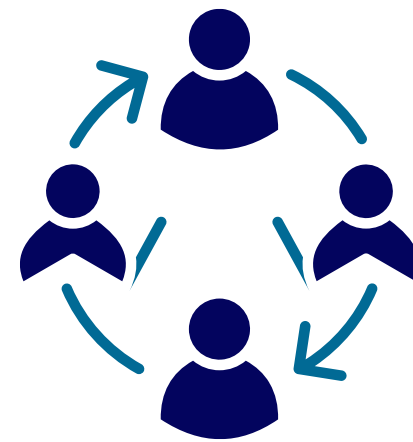
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Background and Motivation



Understand customer
heterogeneity



Reveal hidden patterns
and relationships



Support effective
segmentation and
decision-making

Objective

01

To examine the relationship between customer income and purchasing behavior across different product categories.

02

To analyze how household characteristics, particularly the presence of children, influence household spending patterns.

03

To identify distinct customer segments using Principal Component Analysis (PCA) and K-Means clustering based on income, household characteristics, and purchasing behavior.

Dataset Overview

Here is the dataset from Kaggle named:
Customer Personality Analysis.

	ID	Year_Birth	Education	Marital_Status	Income	Kidhome	Teenhome	Dt_Customer	Recen
0	5524	1957	Graduation	Single	58138.0	0	0	2012-09-04	
1	2174	1954	Graduation	Single	46344.0	1	1	2014-03-08	
2	4141	1965	Graduation	Together	71613.0	0	0	2013-08-21	
3	6182	1984	Graduation	Together	26646.0	1	0	2014-02-10	
4	5324	1981	PhD	Married	58293.0	1	0	2014-01-19	

5 rows × 29 columns

There are 2240 rows and 29 columns.

Customer ID

- Year_Birth
- Education
- Marital_Status
 - Income
 - Kidhome
 - Teenhome
- Dt_Customer
- Complain
- Recency

Source

- NumDealsPurchases
- NumWebPurchases
- NumCatalogPurchases
- NumStorePurchases s
- NumWebVisitsMonth

Product

- MntWines
- MntFruits
- MntMeatProducts
- MntFishProduct
- MntSweetProducts
- MntGoldProds

Promotion

- AcceptedCmp1
- AcceptedCmp2
- AcceptedCmp3
- AcceptedCmp4
- AcceptedCmp5

Data Preprocessing

There are several steps in data preprocessing as following below:

01

**Check and
transform datatype**

02

Missing value

03

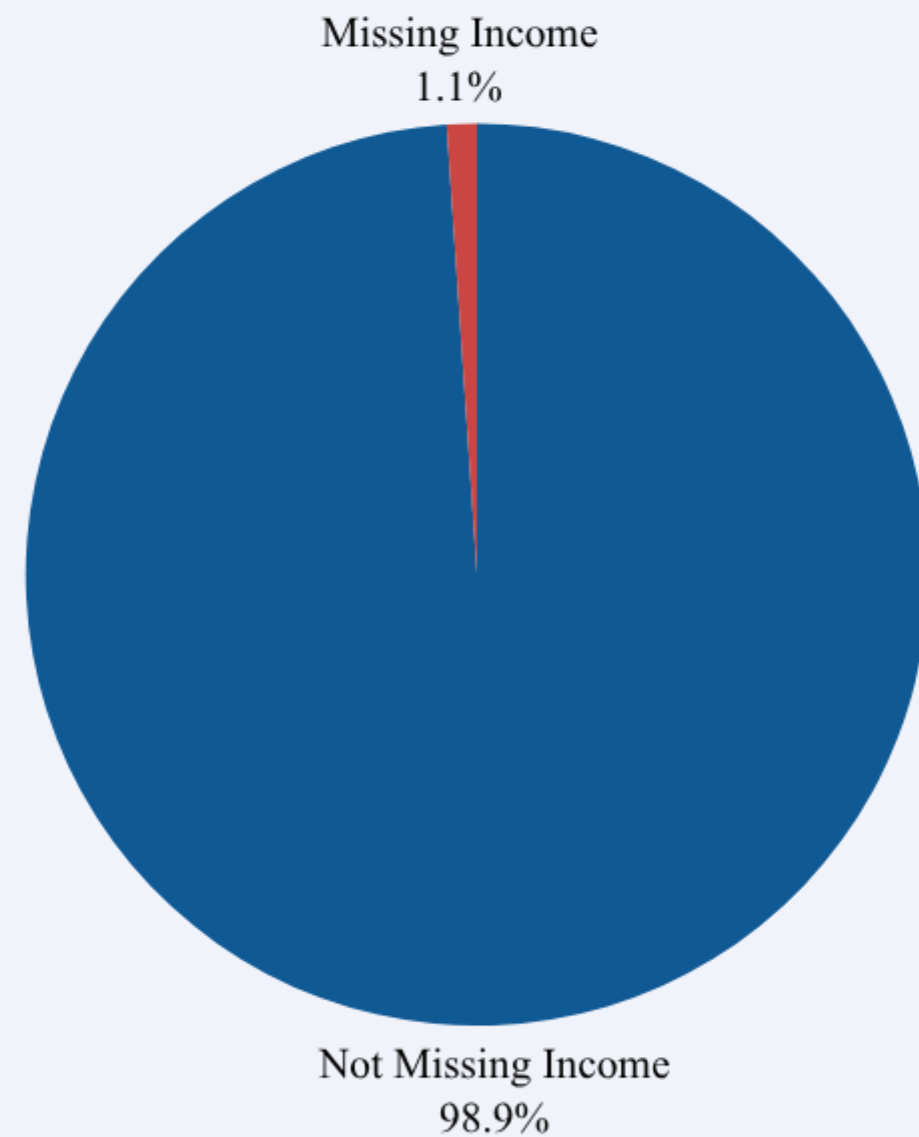
Duplication

04

Outliers

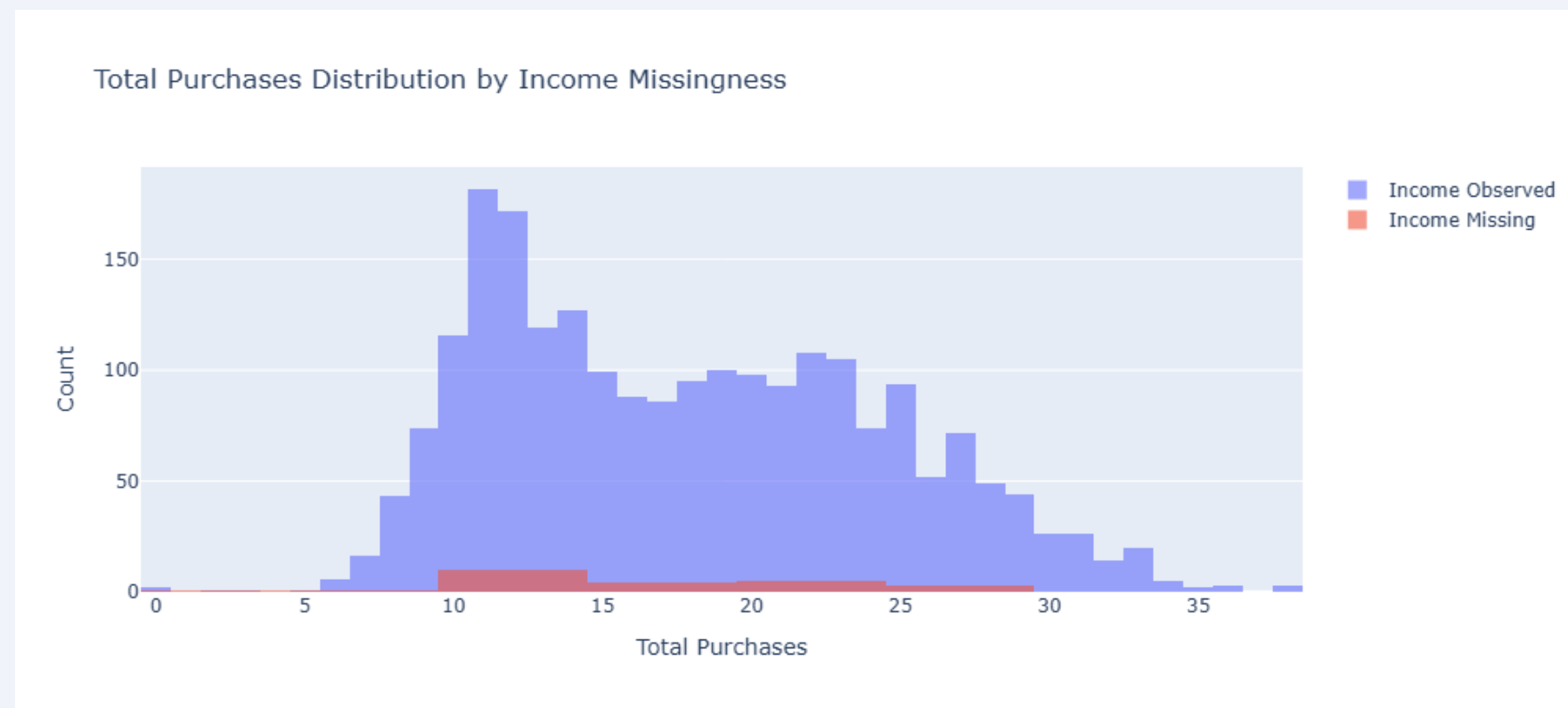
Handle missing value:

Percentage of Missing and Non-Missing Income



In order to be safe we check whether this missing value related or impact to another columns or not.

- **Distribution check:** Shape & range unchanged after dropping missing values
- **Assumption:** Missing Completely at Random (**MCAR**)
- **Implication:** Probability of missingness unrelated to data values



Data Preprocessing

Outlier Detection

Technique used: **IQR**

- $IQR = Q3 - Q1$

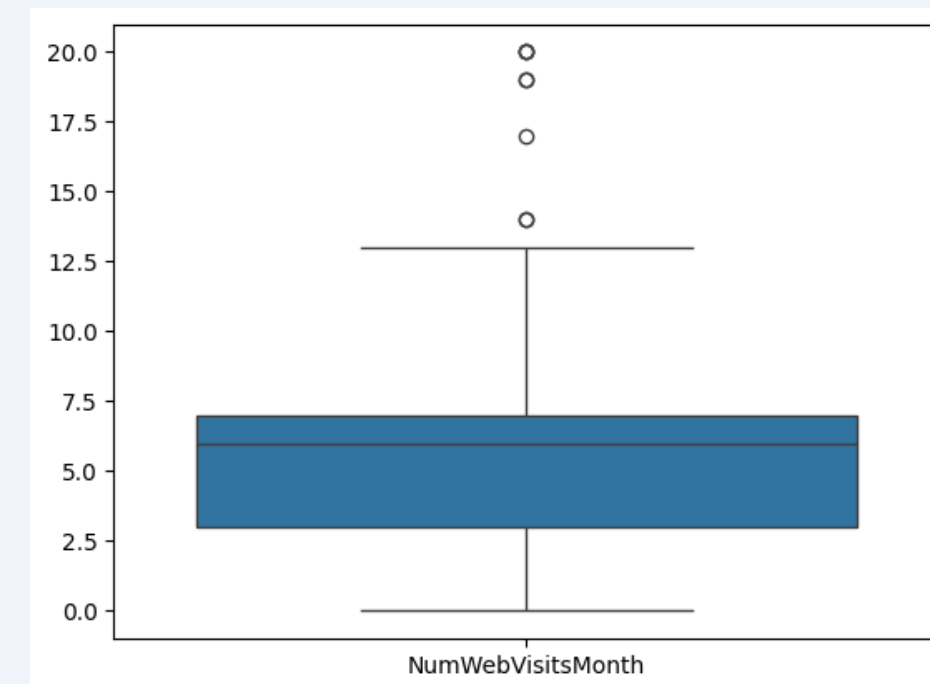
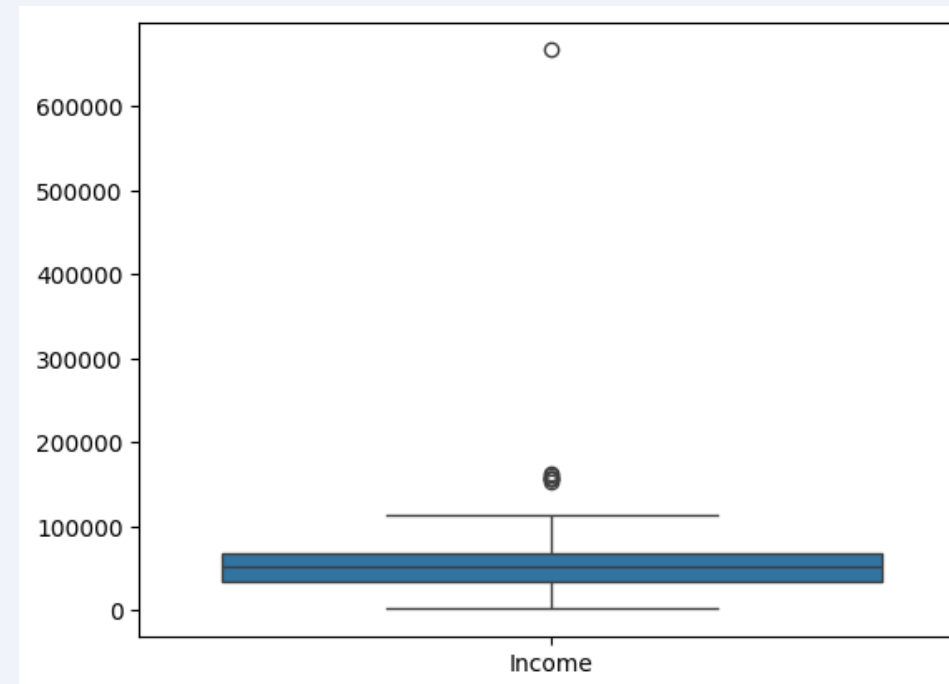
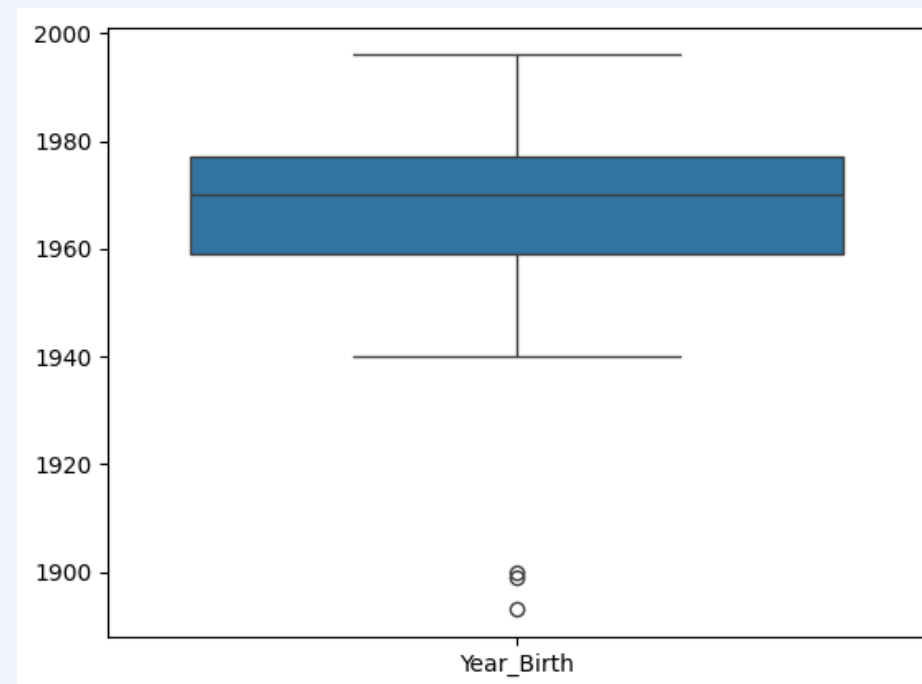
Q1 = 25th percentile

Q3 = 75th percentile

Rule for Outliers

- Lower bound = $Q1 - 1.5 \times IQR$
- Upper bound = $Q3 + 1.5 \times IQR$

Any value outside these bounds is considered an outlier.

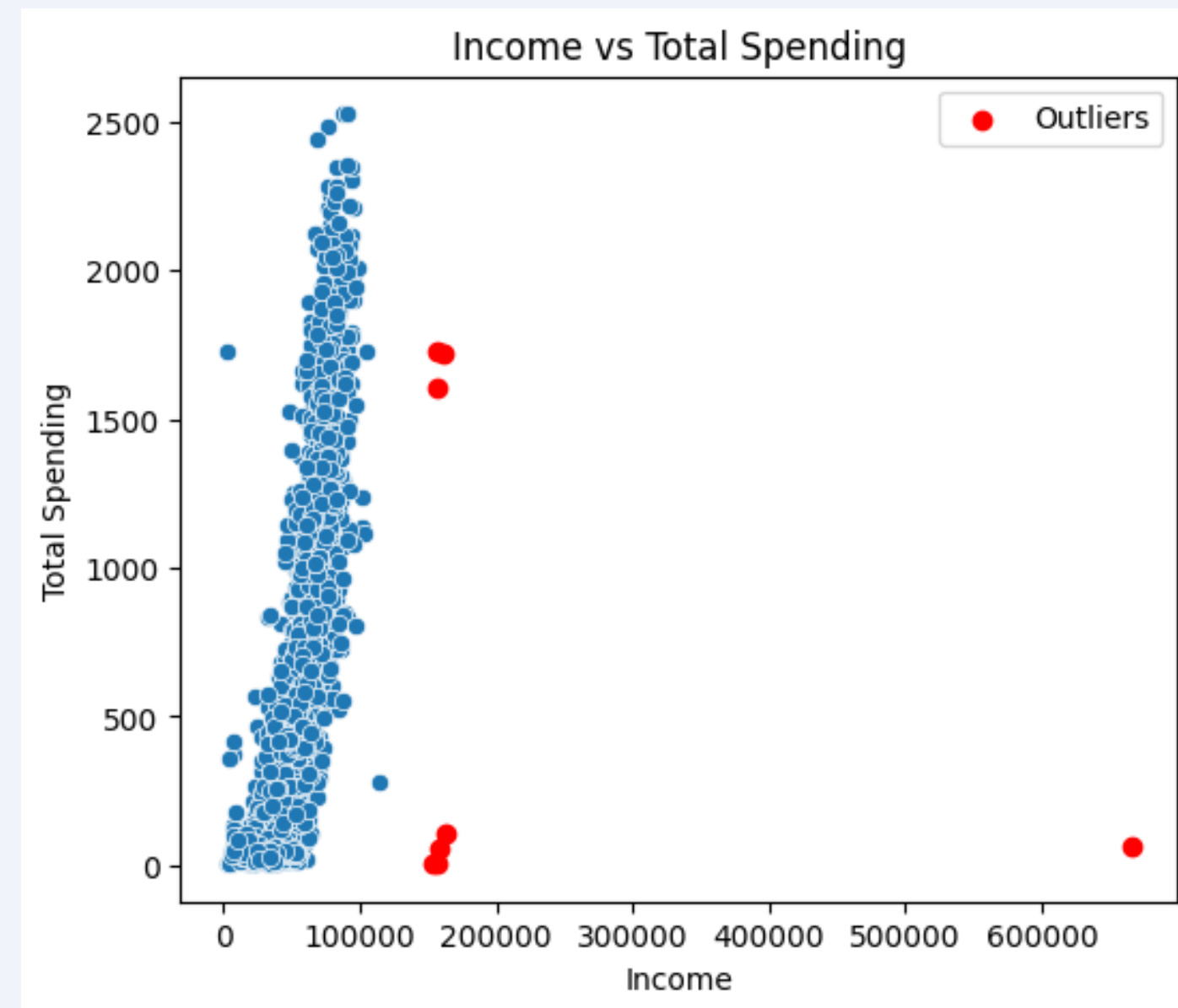
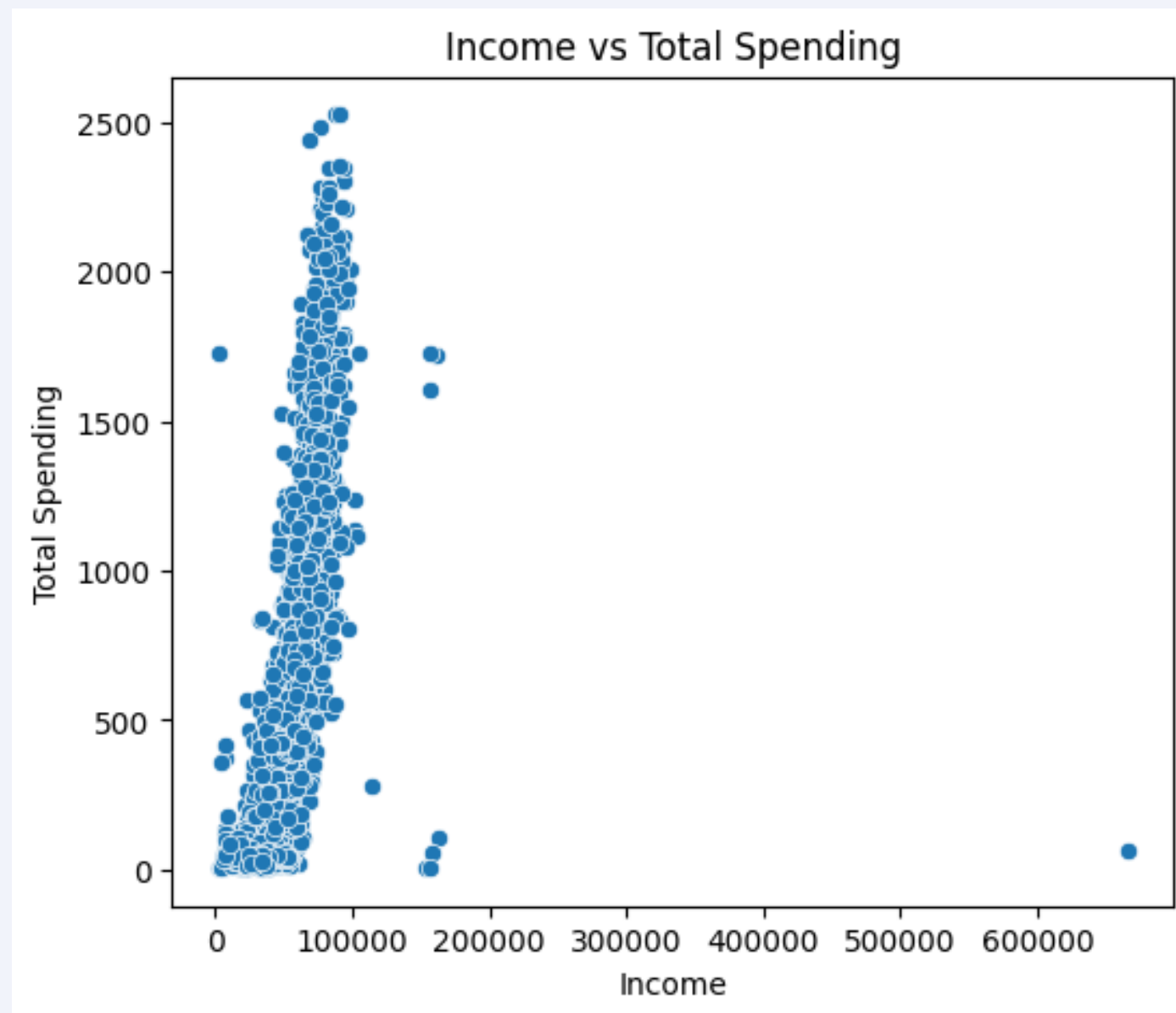


- Less than 1% outlier (Year_Birth: 0.13%, Income: 0.36%, NumWebVisitsMonth: 0.36%)

Data Preprocessing

Outlier Detection

- Outlier more than 1 % to 15%



Feature Engineering

- **Household Structure:**

Combined Kidhome and Teenhome into Kids, and created Kids categories (No Kids, One Kid, Two+, Three+).

- **Spending Behavior:**

Aggregated product-level spending into Total_Spending to capture overall purchase intensity and reduce redundancy.

- **Purchase Activity:**

Combined purchase channels into Total_Purchases and created Deal_Intensity to measure price sensitivity.

- **Data Transformation:**

Applied log transformation for skewed spending variables (visual clarity) and standardization before PCA and clustering.

Why Feature Engineering?

- Statistical evidence: High correlations among similar variables
- Methodological need: PCA and clustering perform better with fewer, conceptually coherent features.
- Research relevance: reflect the study's objectives and customer behavior.

- Univariate Analysis

Descriptive statistics of numerical columns

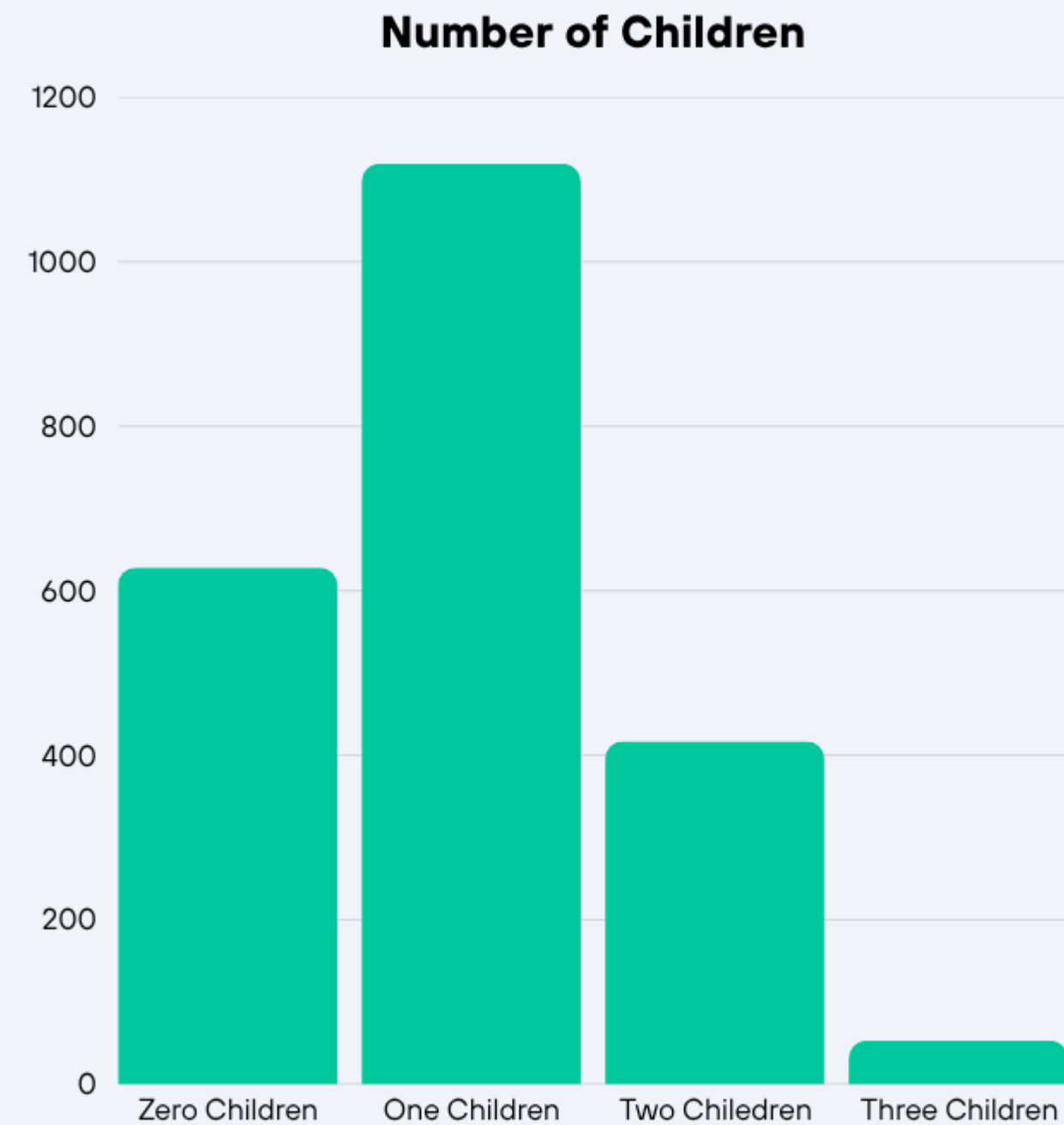
	Income	Age	kids	total_purchases	expenses	total_accepted_campaigns	total_purchases_log	expenses_log
count	2217	2217	2217	2217	2217	2217	2217	2217
mean	51809.613343	45.107352	0.952639	12.567885	607.711773	0.299955	2.446105	5.630766
std	20377.133953	11.698615	0.751661	7.142377	601.752444	0.680946	0.594688	1.461541
min	2447	18	0	1	8	0	0.693147	2.197225
25%	35688	37	0	6	69	0	1.94591	4.248495
50%	51717	44	1	12	400	0	2.564949	5.993961
75%	68126	55	1	18	1048	0	2.944439	6.955593
max	10 5471	74	3	32	2525	4	3.496508	7.834392

- Univariate Analysis

Distribution of numerical columns

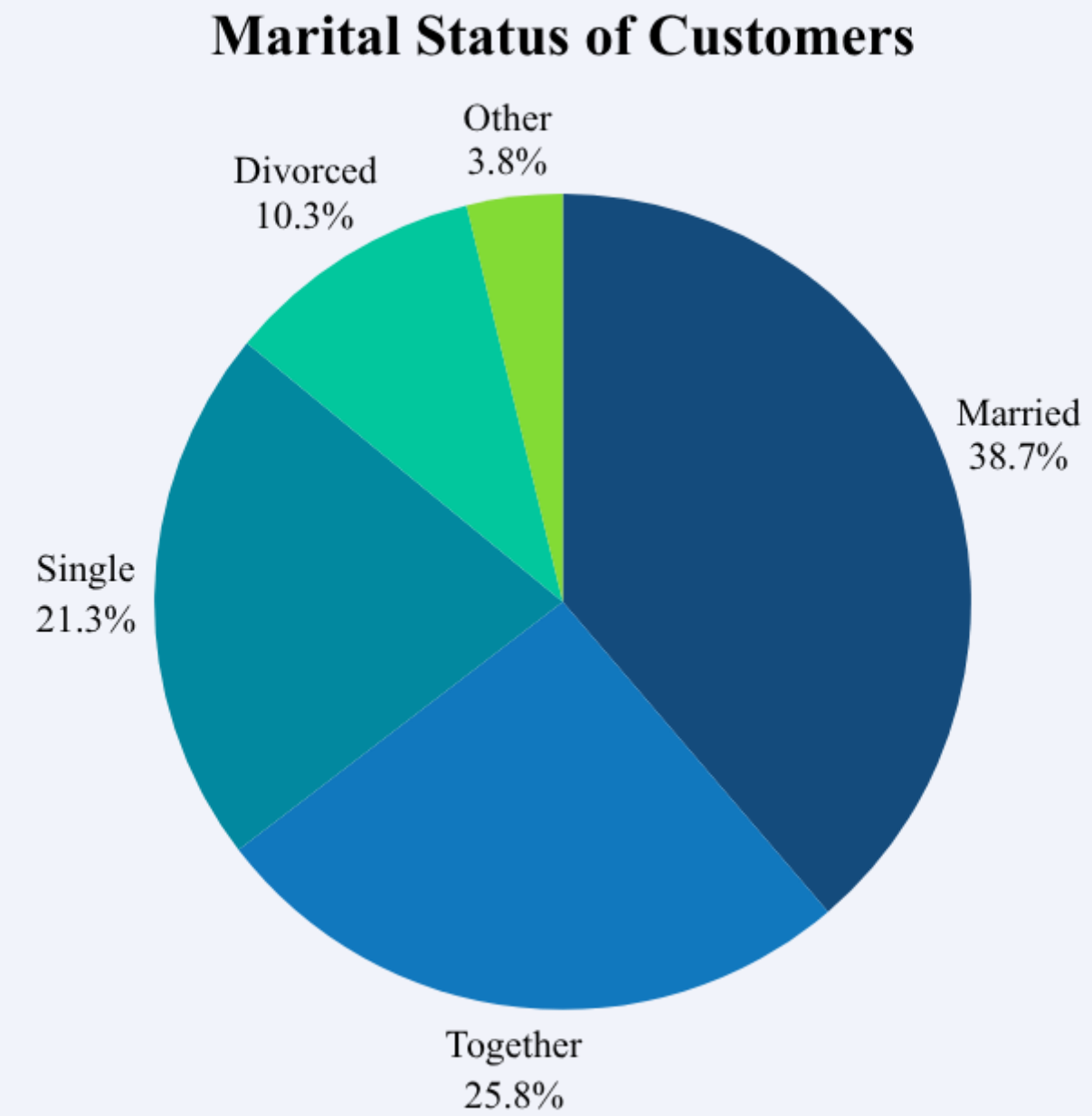
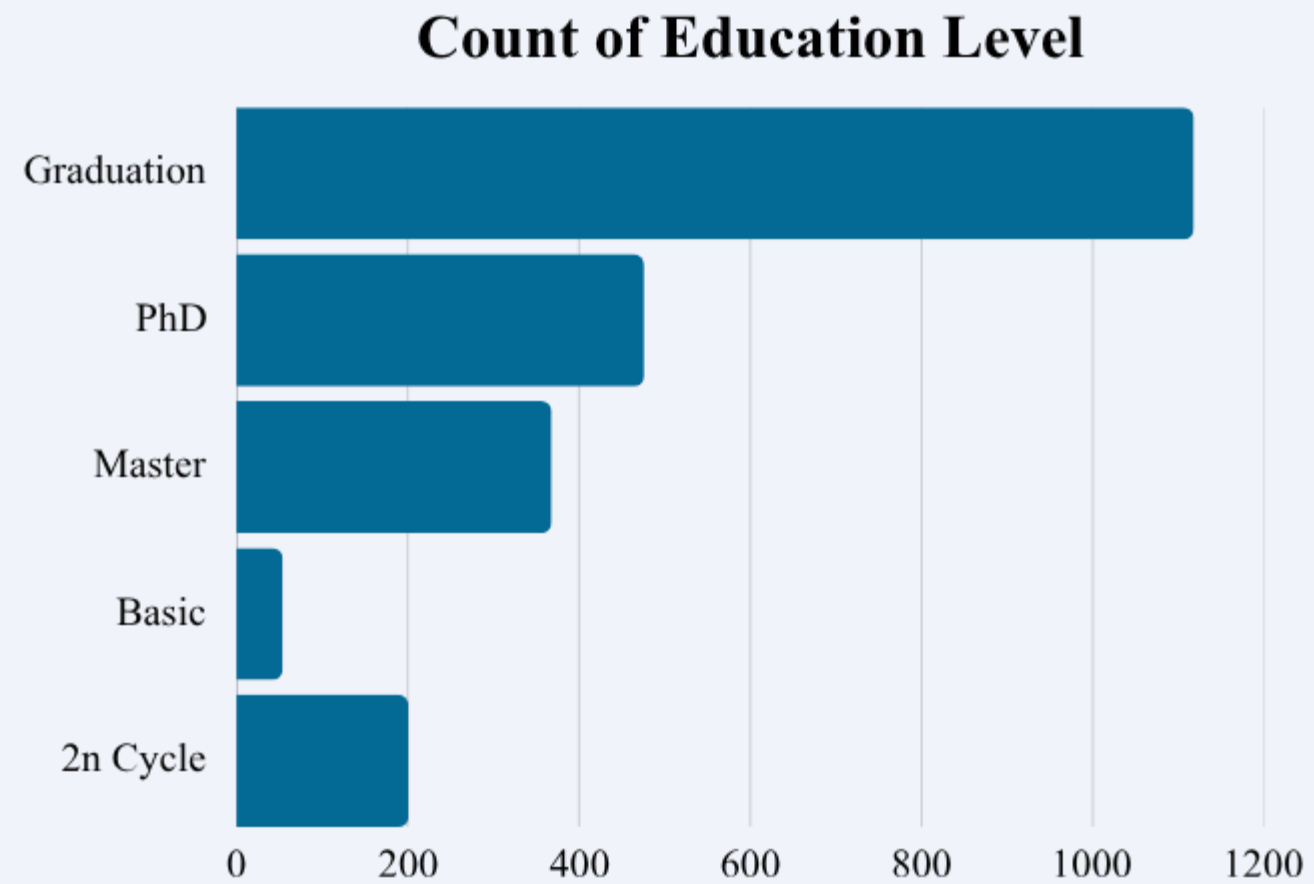


Distribution of categorical column



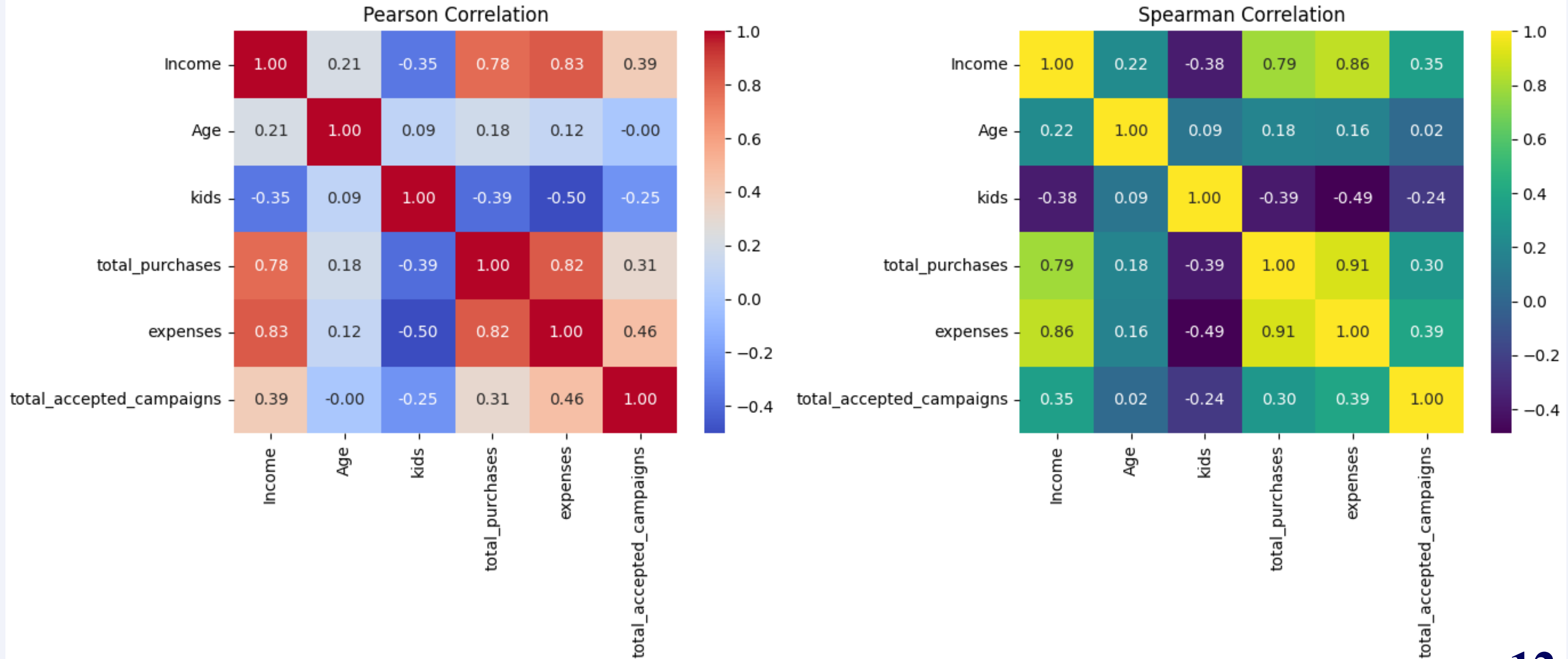
- Univariate Analysis

Proportion of Categorical Columns



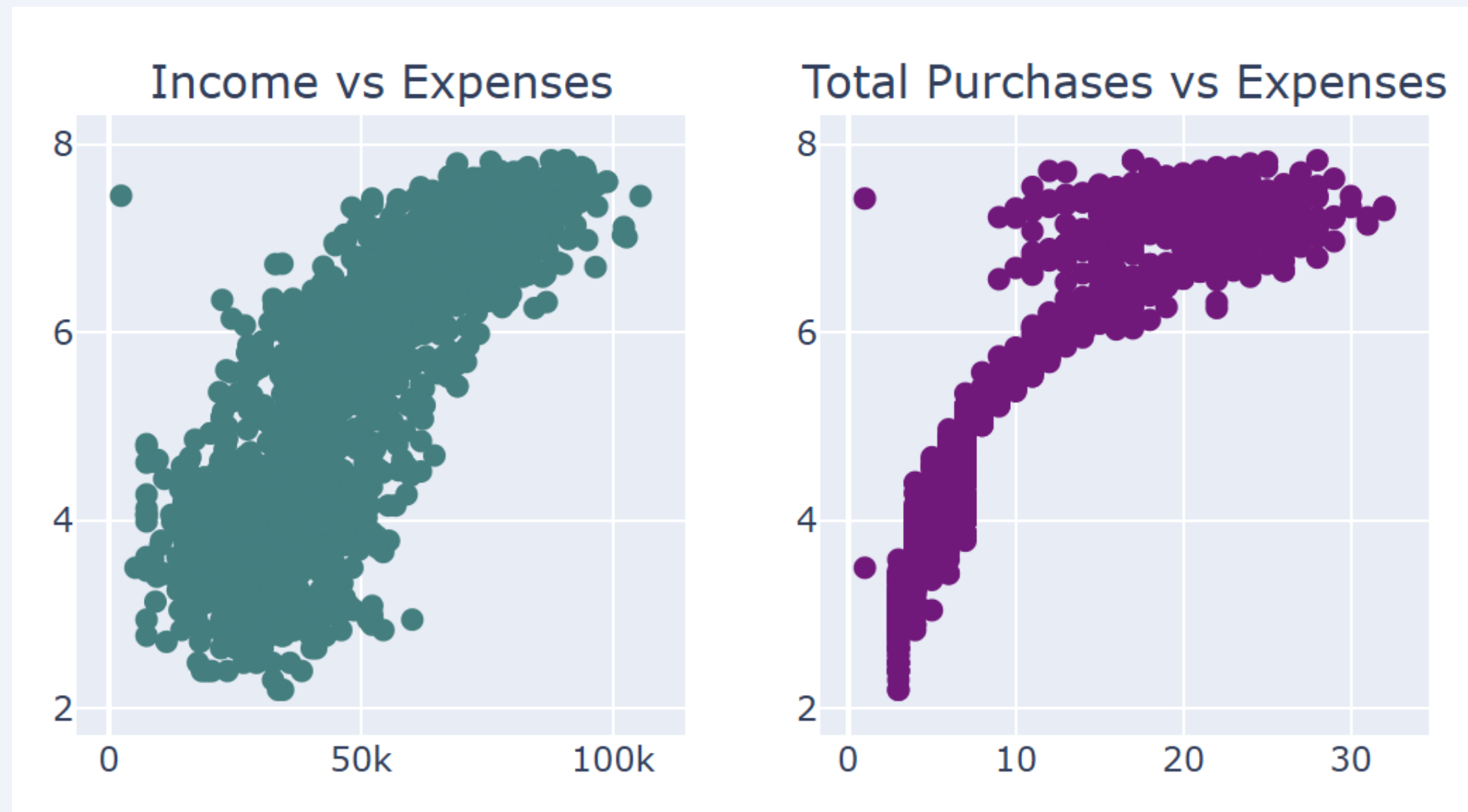
- Bivariate Analysis

Correlation matrix of numerical columns



- Bivariate Analysis

Quantitative vs Quantitative



- Scatterplot shows relation of quantitative pairs
- **Income vs Expenses**
 - A strong positive relationship exists between income and expenses.
 - As income increases, customer spending generally increases, though with greater variability at higher income levels.
- **Total Purchases vs Expenses**
 - A clear positive association, suggesting purchase frequency is a strong predictor of total expenditure.
 - Purchase frequency strongly influences total customer spending.

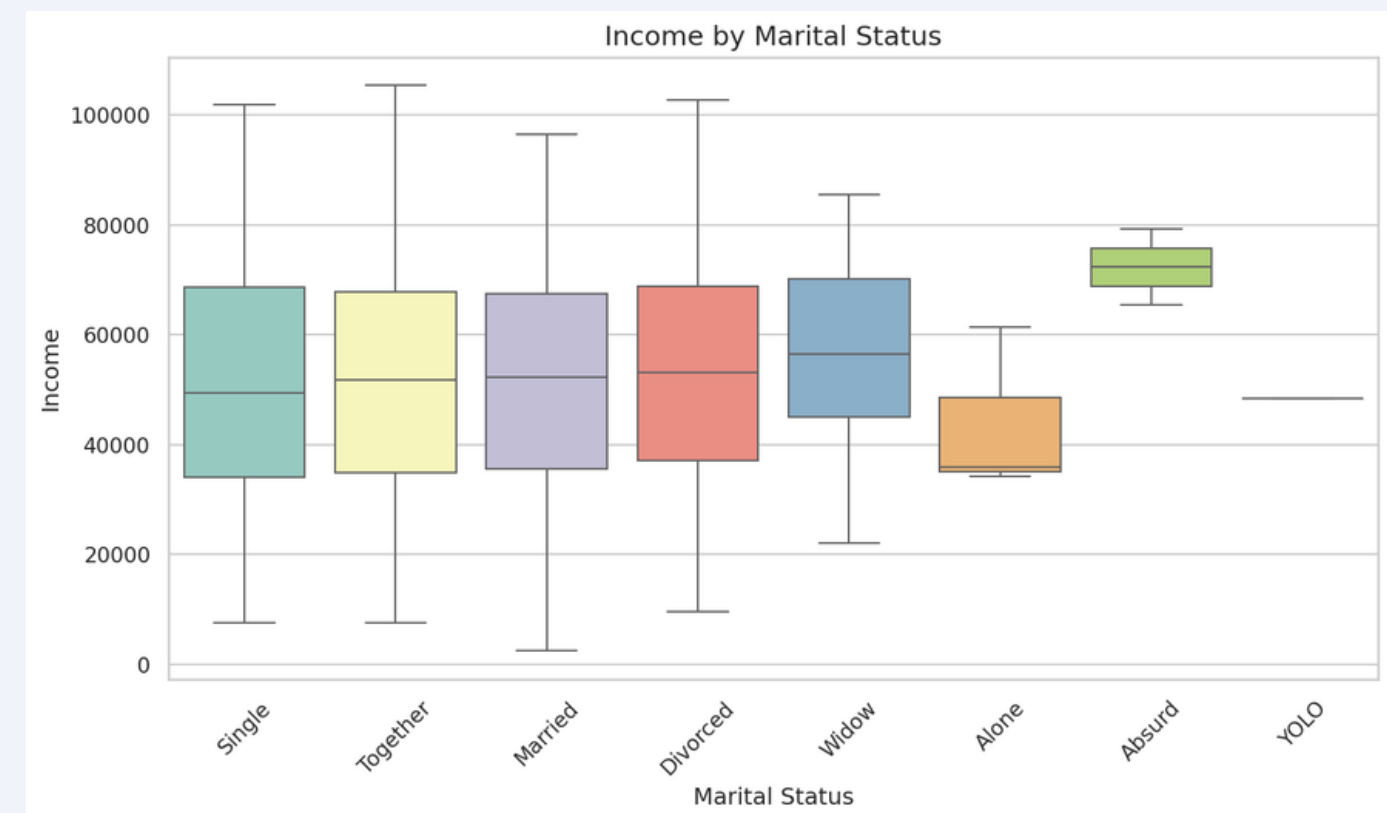
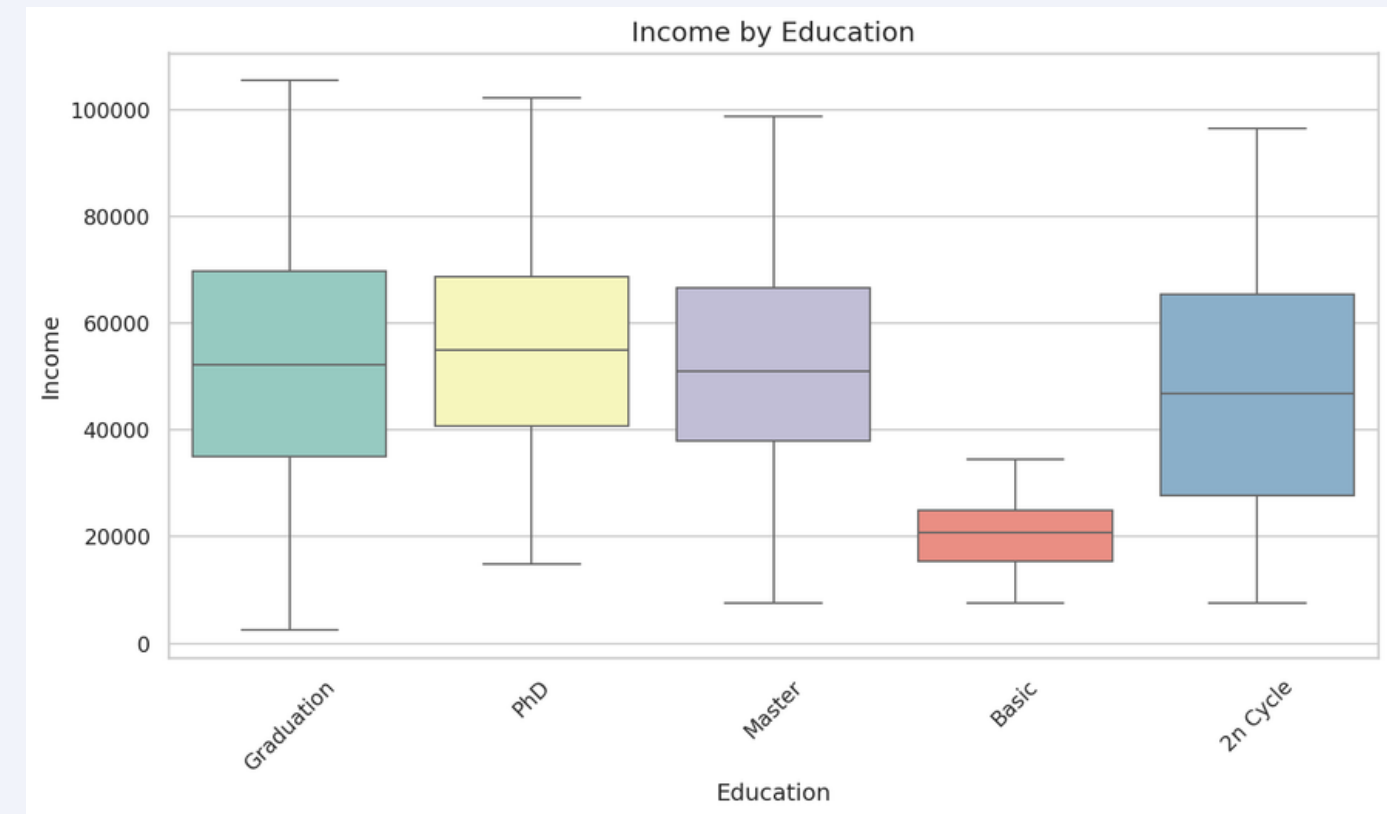
- Bivariate Analysis

Quantitative vs Qualitative

Higher education is generally **associated** with **higher income**, though income distributions overlap across groups, reflecting heterogeneity within education levels.

Partnered or previously partnered groups (Married, Together, Divorced, Widow) have **higher median incomes** than Single and Alone.

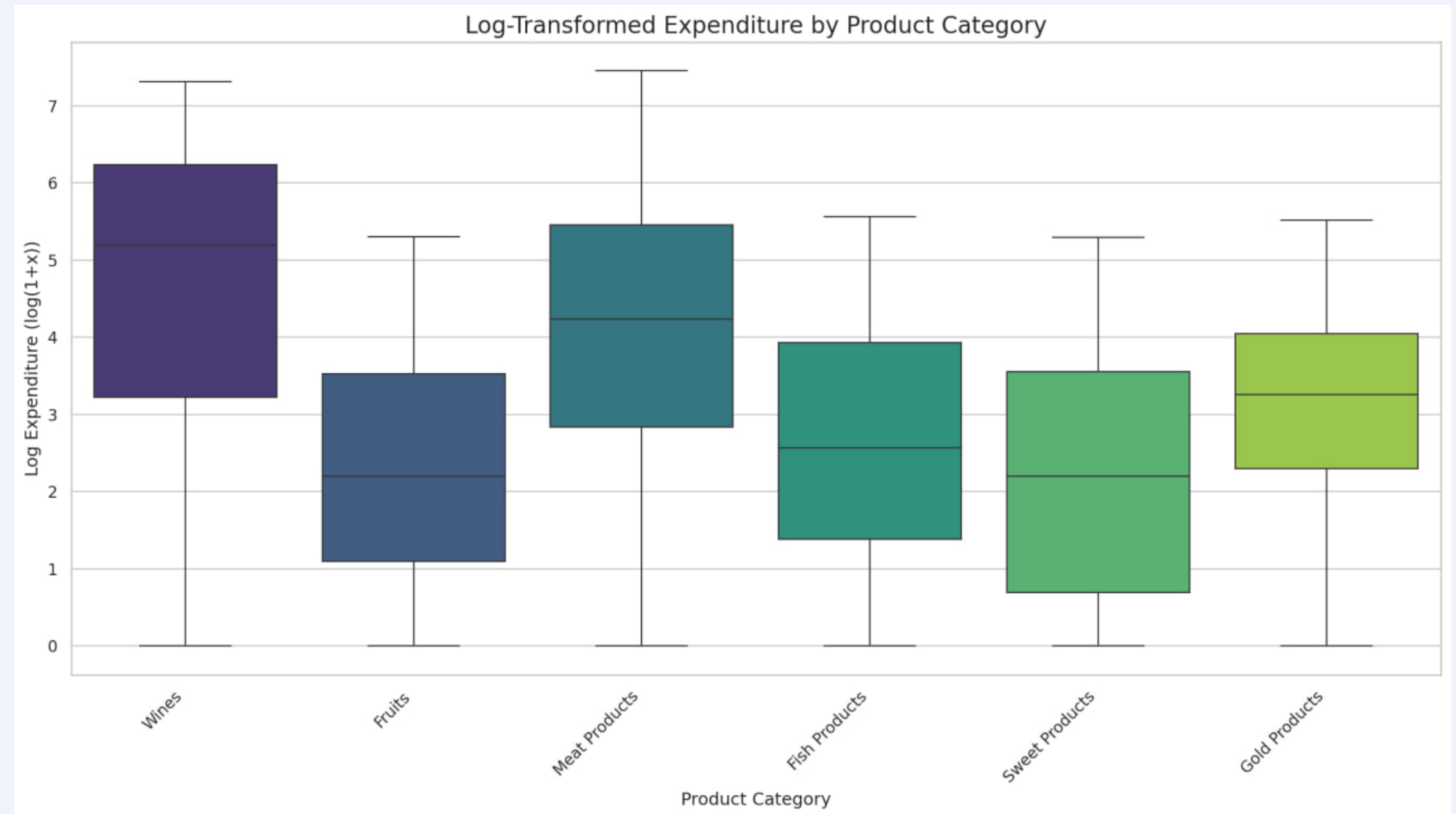
Rare categories (Absurd, YOLO) show extreme values.



- Bivariate Analysis

- Quantitative vs Qualitative

The log plot highlights relative spending behavior across product categories while minimizing the influence of extreme values.

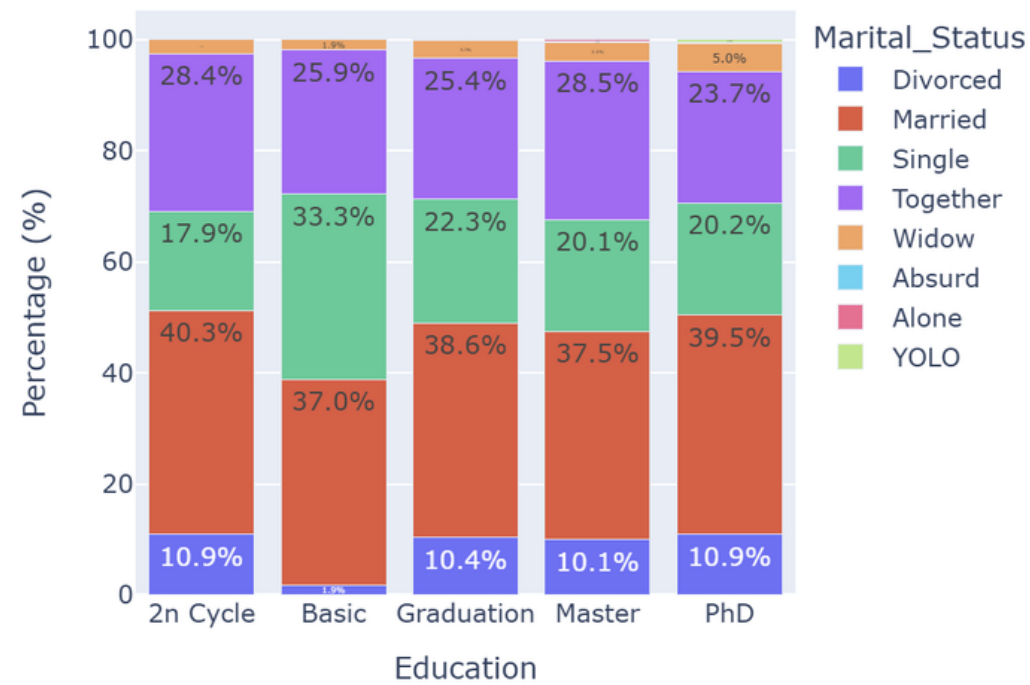


- **Wines** and **Meat Products** still show the **highest typical expenditure**, indicating they are the primary spending categories.
- **Fruits** and **Sweet Products** have consistently **lower spending**, while **Fish** and **Gold Products** **fall in between**, confirming stable relative differences across categories.

- Bivariate Analysis

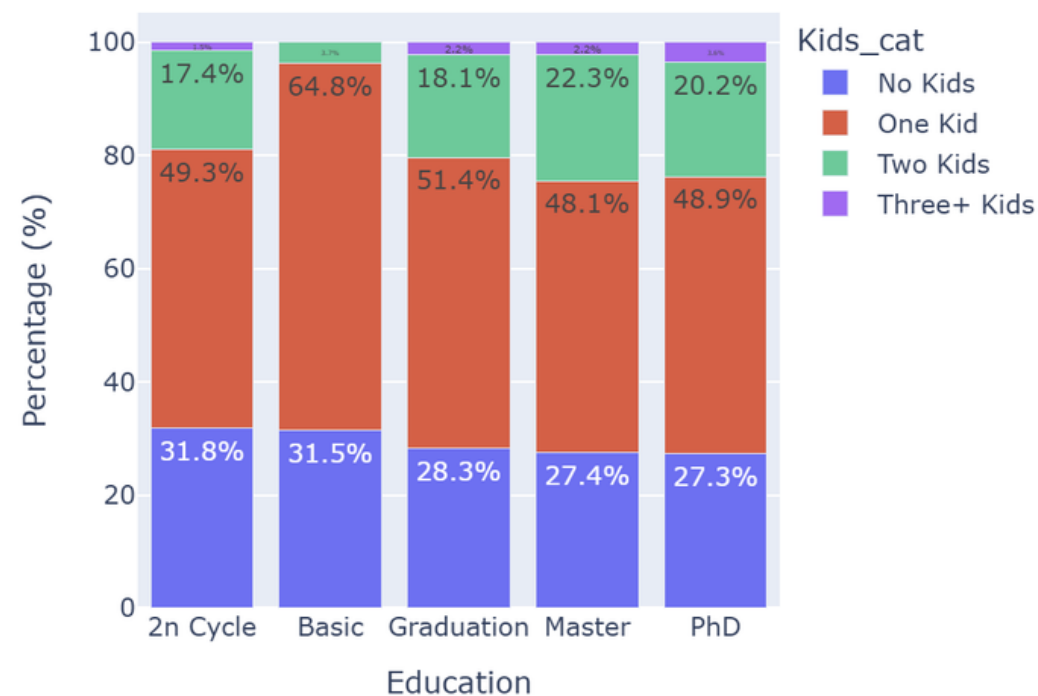
Qualitative vs Qualitative

Education Level by Marital Status



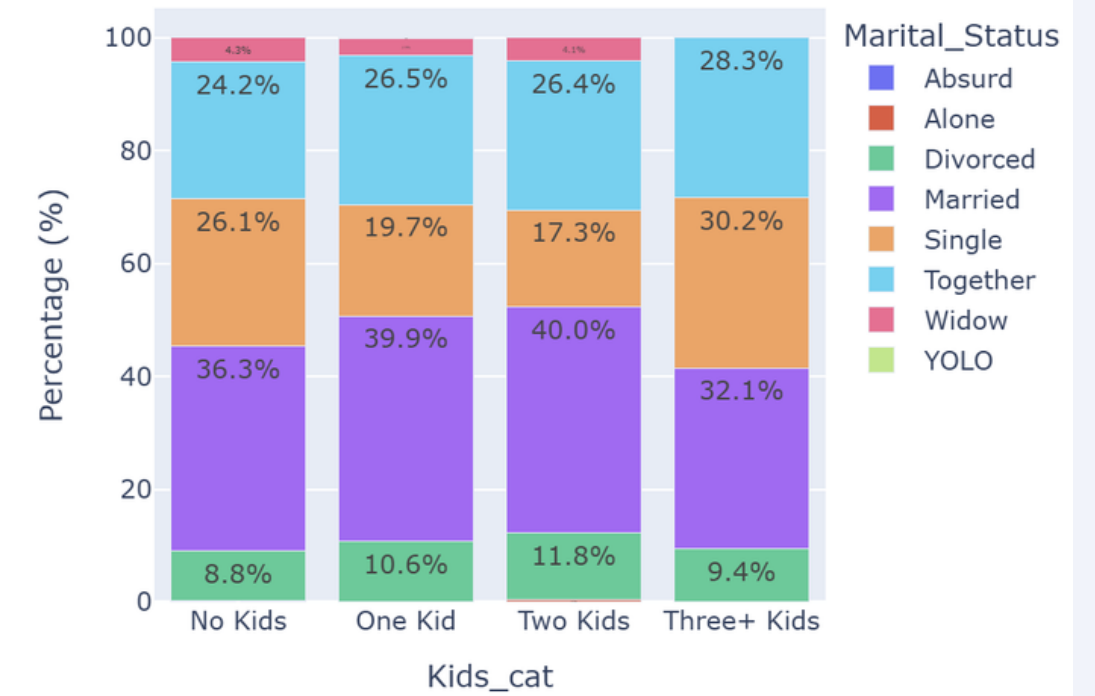
The visual similarity across bars

Education Level by Number of Children



no strong structural shift

Marital Status by Number of Children



Clear pattern difference exists

- Bivariate Analysis - Correspondence Analysis (CA)

**Statistical test****Education Level vs Marital Status**

- H0: Education level and marital status are independent.
- H1: Education level and marital status are not independent.

Chi-square test

chi2 statistic = 27.9250
dof = 28
p-value = 0.468429

Education vs Kids

- H_0: Education level and the presence of children are independent.
- H_1: Education level and the presence of children are associated.

Chi-square test

chi2 statistic = 19.2776
dof = 12
p-value = 0.0820449

Since, both studies provided the p-value is greater than the significance level of 0.05, the null hypothesis cannot be rejected.

Interpretation

- This result indicates that there is **no statistically significant association** between **education level** and **marital status**.
- This indicates that there is **no statistically significant association** between **education level** and the **number of children**.

- Bivariate Analysis - Correspondence Analysis (CA)

Marital Status vs Kids

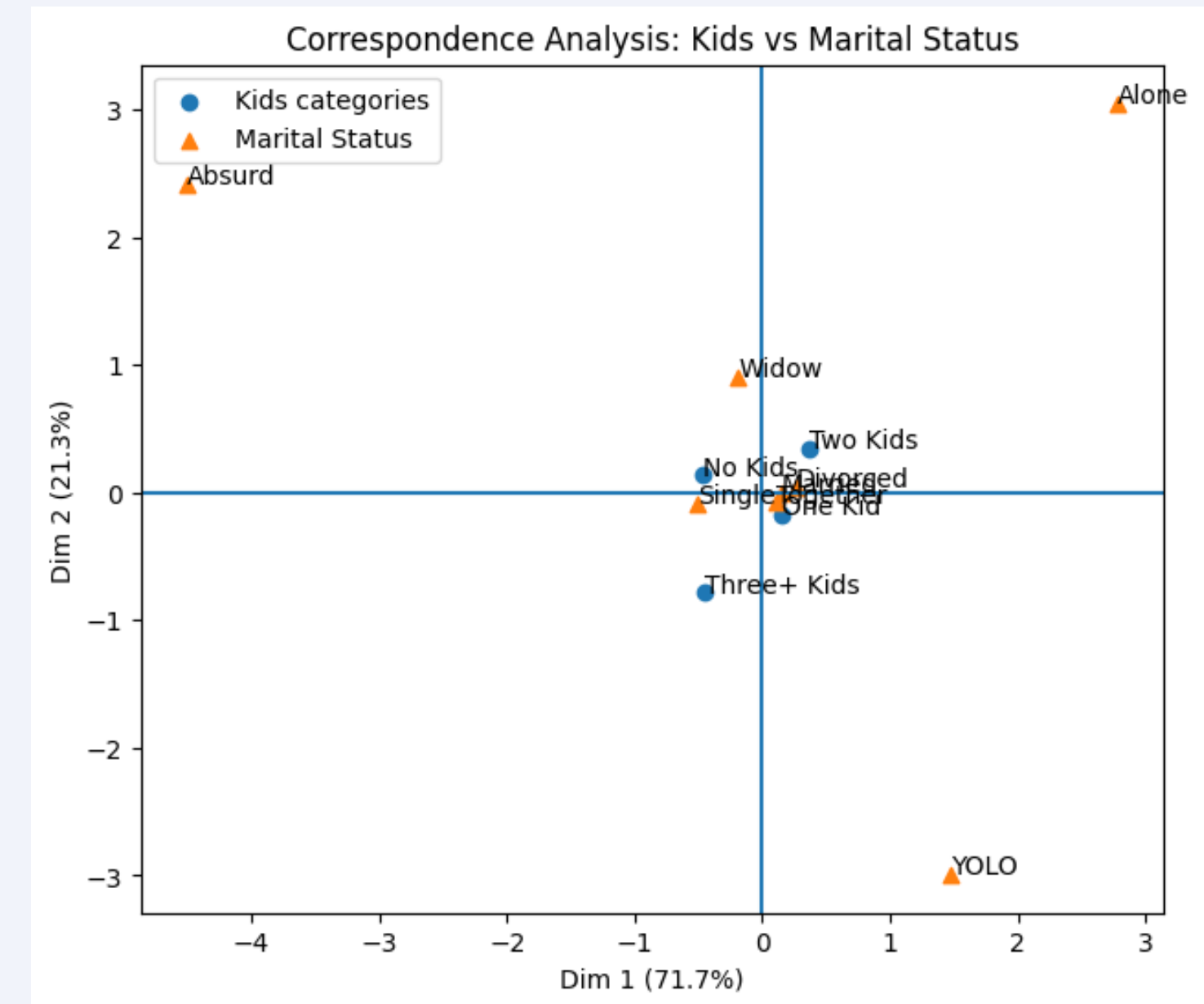
- H0: Kids category and marital status are independent.
- H1: Kids category and marital status are not independent

Chi-square test

chi2 statistic = 35.0164
dof = 21
p-value = **0.028**

Since $p\text{-value} < \alpha = 0.05$, we can reject H0

Kids and Marital_Status are associated



Interpretation

- Married** and **Together** are closely associated with **Two Kids** / **Three Kids**, indicating family-oriented households.
- Single** and **Widow** are located near **No Kids** / **One Kid**, suggesting childless or smaller households.
- Rare categories (YOLO, Alone, Absurd) are far from the origin, showing distinct but uncommon profiles.

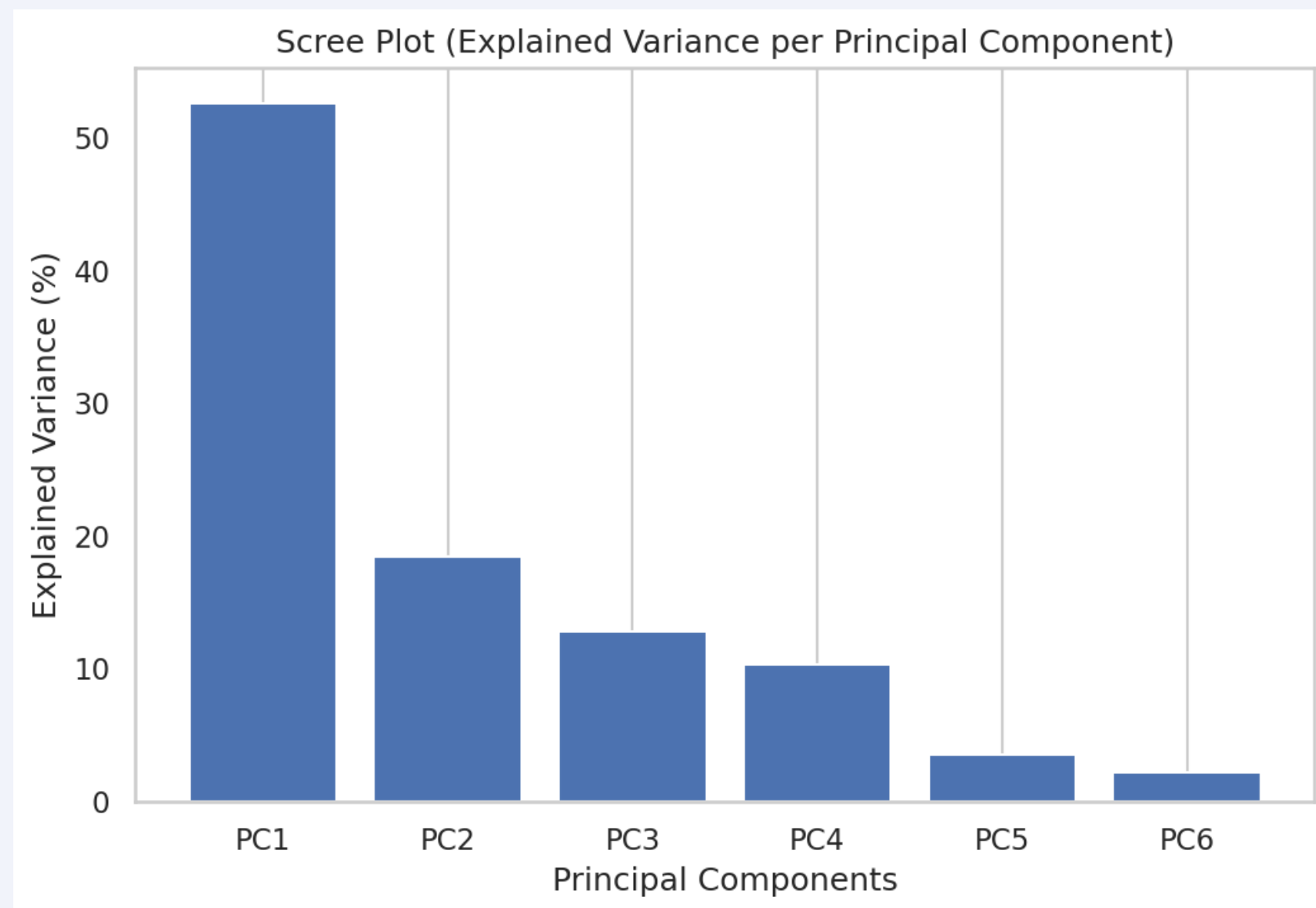
- Multivariate Analysis - PCA

PCA was applied to standardized numerical variables representing **customer demographics** and **purchasing behavior**, including income, age, number of children, total purchases, expenses, and campaign acceptance.

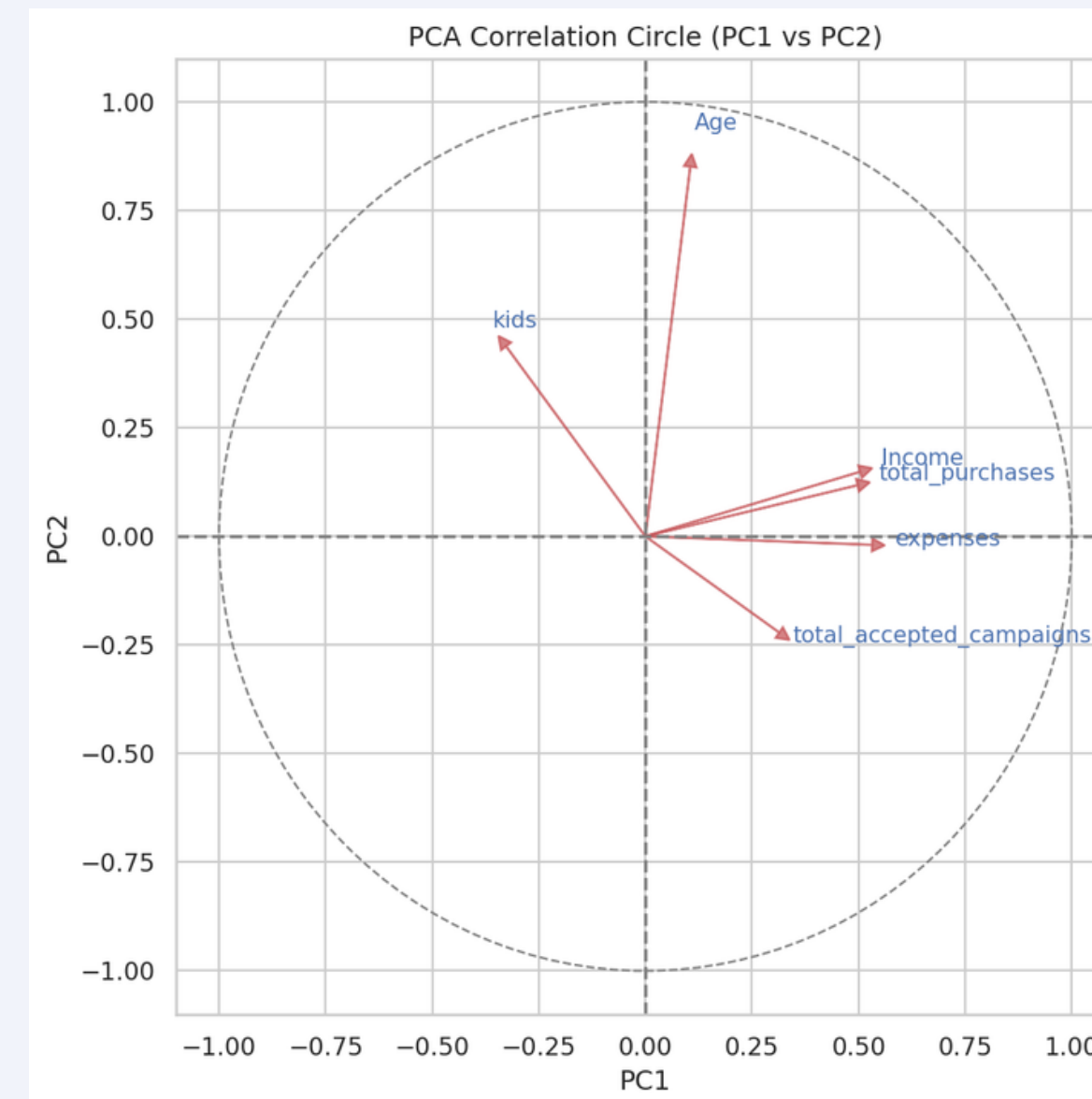
	PC1	PC2	PC3	PC4	PC5	PC6
Income	0.503511	0.14856	-0.036631	0.272337	0.718509	-0.364201
Age	0.10513	0.850605	0.0474	-0.510404	-0.034017	0.038771
Kids	-0.326691	0.437268	0.417095	0.713139	-0.043122	0.132949
Total purchase	0.497488	0.117579	-0.193467	0.297611	-0.686347	-0.376305
expenses	0.531735	-0.019591	-0.059101	0.134724	-0.00001	0.833805

- **PC1** is primarily driven by Income, expenses, and total_purchases, reflecting **customer purchasing power**.
- **PC2** is influenced by Age and kids, capturing **demographic differences**.

- Multivariate Analysis - PCA



- **PC1** captures around 50 %
- **PC2** captures around approximately 20%

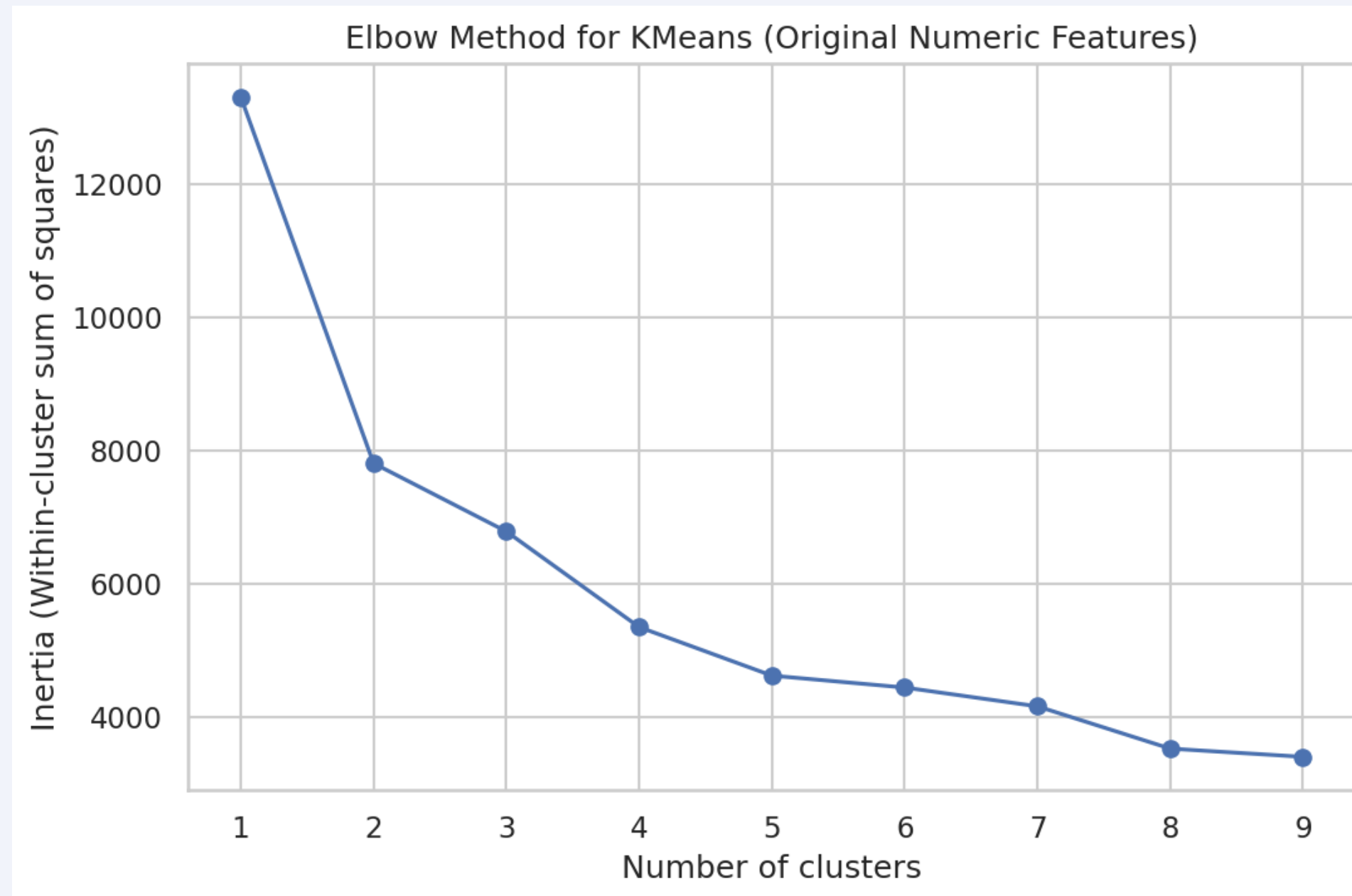


- **PC1** reflecting **customer purchasing**
- **PC2** capturing **demographic differences**.

K-mean Clustering

Features for Clustering (Income, Age, kids, total_purchase, and expense)

- **Elow Technique:** Identify number of K-Cluster



- **Silhouette Score**

Nearly 1 is very good, 0.25 to 50 is acceptable, and otherwise is poor

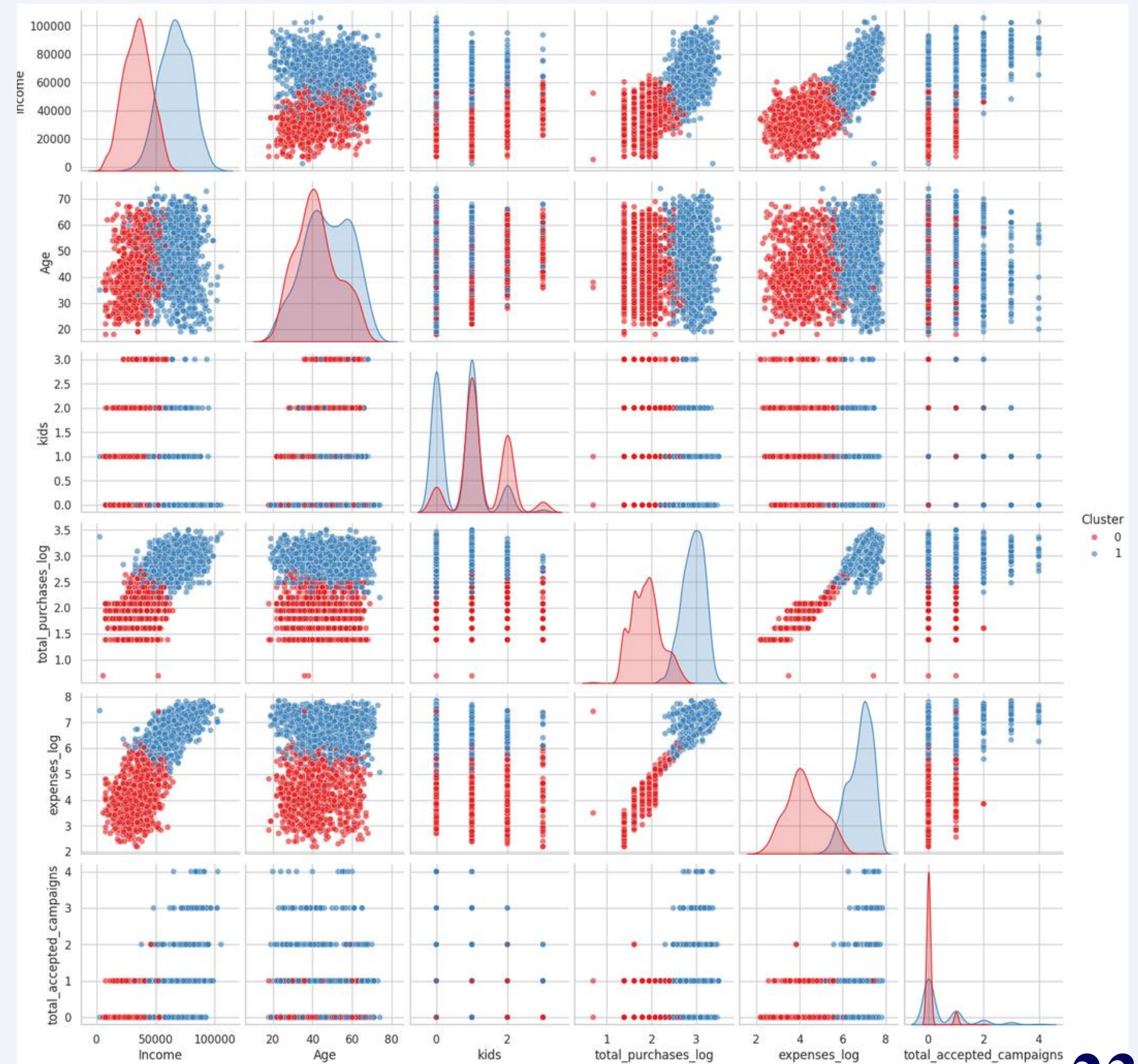
K-Cluster	Silhouette score
K= 2	0.368
K = 3	0.285
K = 4	0.298
K= 5	0.269
K = 6	0.248
K = 7	0.242
K = 8	0.275
K = 9	0.257

K-mean Clustering

- **Cluster 0:** Low-Income, young, low expenses
- **Cluster 1:** High-Income, old, high expenses

Cluster (K=2)		0	1
Income	Min	5305.00	34294.17
	Mean	34294.17	66573.25
	max	34294.17	66573.25
Age	Min	18.00	19.00
	Mean	42.24	47.53
	max	69.00	74.00
Kids	Min	0.00	0.00
	Mean	1.29	0.67
	max	3.00	3.00
Total_purchase	Min	1.00	8.00
	Mean	5.90	18.19
	max	14.00	32.00
expense	Min	8.00	157
	Mean	98.61	1036.83
	max	1679.00	2525.00

K-mean Clustering



Keys Finding

Correlation

- **Income vs Expense:** Positive correlation
The higher income is more likely to high expense
- **Income vs Total Purchase:** Positive correlation
The higher income is more likely to high purchase
- **Income vs education level** has associated, Clear differences across categories, especially between Basic and 2n-cycle education.

Future Work

- Corporate Data with Temporal Dimension
- Improving K-Clustering for Interpretability
- Applying Supervised Machine Learning (classification/regression models)

Demographic Associations

- **Marital Status** vs **Kids:** has associated each other. Single person more likely to have no kids, while together and marital have two or three kids.

K-mean clustering

- **Cluster 0** represented young people, low income, and in average has one kids (young family).
- **Cluster 1** represented old people, high income, and more likely to have no kids (Affluent Empty Nesters).

References

Jolliffe, I. T., & Cadima, J. (2016). Principal component analysis: A review and recent developments. Philosophical Transactions of the Royal Society A: Mathematical, Physical and Engineering Sciences, 374(2065), 20150202.

<https://doi.org/10.1098/rsta.2015.0202>

Little & Rubin (2019) describe MCAR as a situation in which missing values occur entirely by chance and are independent of both observed and unobserved data, supporting the use of deletion under these conditions.

[univariate analysis](#)

<https://github.com/Povsundra/EDA-DataScience/blob/main/Project/CA.ipynb>

https://haralick.org/DV/Handbook_of_Data_Visualization.pdf

Thank You !



Introduction

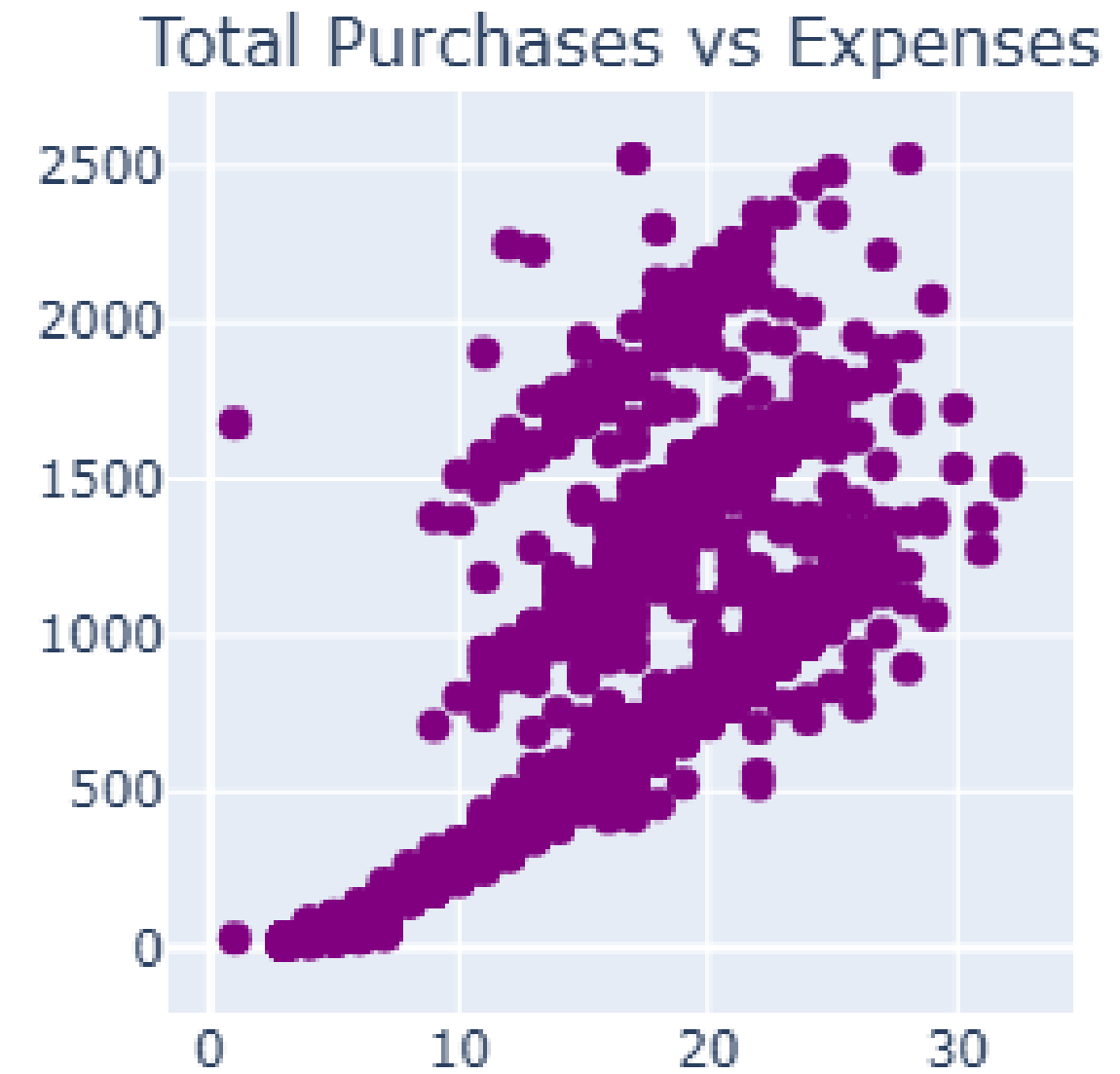
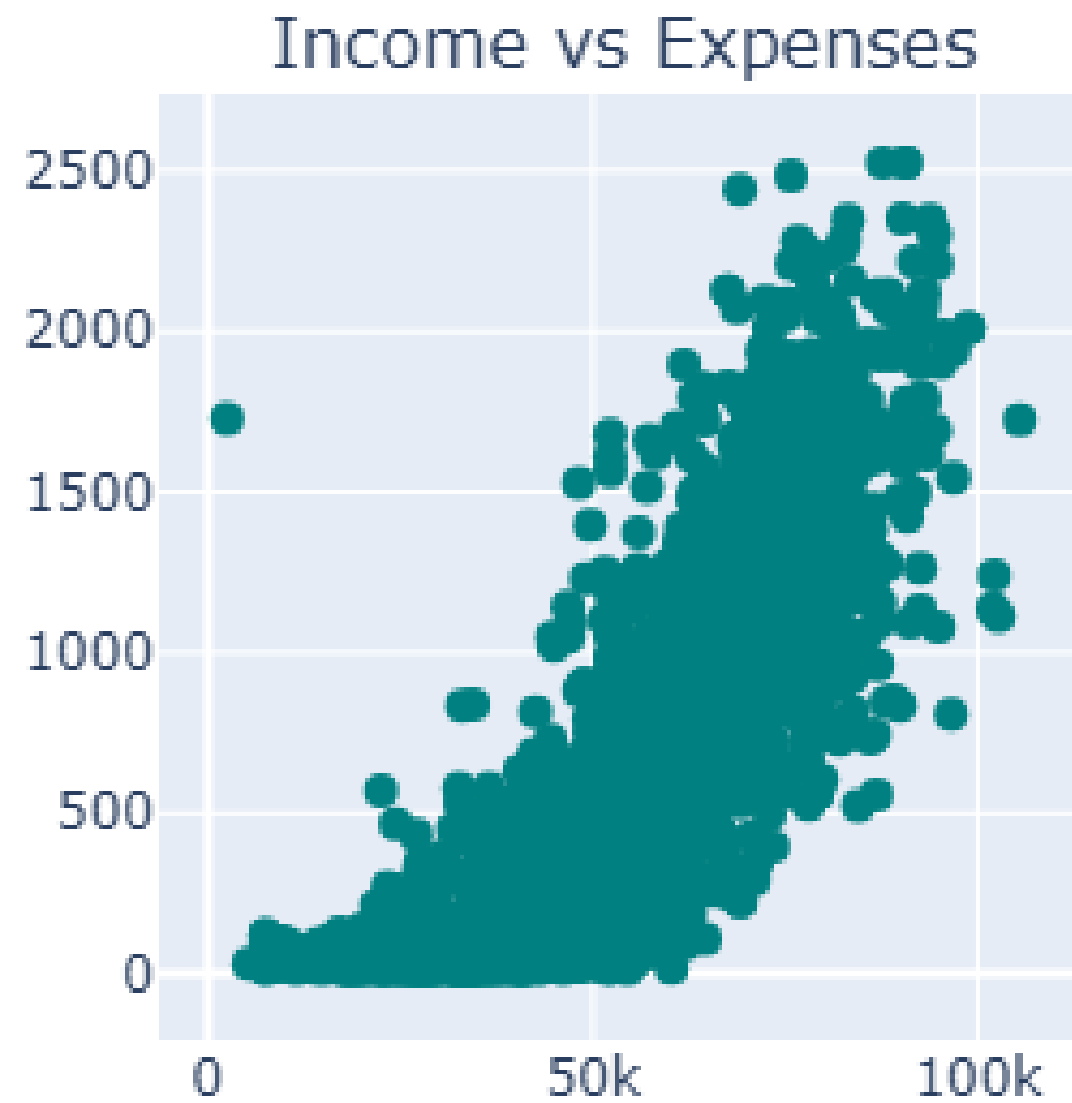
Questions

1. How does customer income relate to purchasing behavior across different product categories?
2. How do household characteristics, especially the presence of children, affect household spending patterns?
3. What distinct customer segments can be identified using PCA and K-Means clustering based on income, household characteristics, and purchasing behavior?

- Bivariate Analysis

Quantitative vs Quantitative

Income and Total Purchases vs Expenses



- Bivariate Analysis

Quantitative vs Quantitative

